tf.fill

View source on GitHub (https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/array_ops.py#L204-L250)

Creates a tensor filled with a scalar value.

```
tf.fill(
    dims, value, name=None, layout=None
)
```

Used in the notebooks

Used in the guide	Used in the tutorials
• Ragged tensors (https://www.tensorflow.org/guide/ragged_tensor) • tf.data: Build TensorFlow input pipelines (https://www.tensorflow.org/guide/data) • Subword tokenizers (https://www.tensorflow.org/text/guide/subwords_tokenizer) • Unicode strings (https://www.tensorflow.org/text/guide/unicode)	• Scalable model compression (https://www.tensorflow.org/tutorials/optimization/compression) • Multiple changepoint detection and Bayesian model selection (https://www.tensorflow.org/probability/examples/Multiple_changepoint_detection_and_Bayesian_model_selection) • Graph-based Neural Structured Learning in TFX (https://www.tensorflow.org/tfx/tutorials/tfx/neural_structured_learning) • Graph regularization for sentiment classification using synthesized graphs (https://www.tensorflow.org/neural_structured_learning/tutorials/graph_keras_lstm_imdb) • Bayesian Gaussian Mixture Model and Hamiltonian MCMC (https://www.tensorflow.org/probability/examples/Bayesian_Gaussian_Mixture_Model)

See also `tf.ones` (https://www.tensorflow.org/api_docs/python/tf/ones), `tf.zeros` (https://www.tensorflow.org/api_docs/python/tf/zeros), `tf.one_hot` (https://www.tensorflow.org/api_docs/python/tf/one_hot), `tf.eye` (https://www.tensorflow.org/api_docs/python/tf/eye).

This operation creates a tensor of shape `dims` and fills it with `value`.

For example:

```
>>> tf.fill([2, 3], 9)
<tf.Tensor: shape=(2, 3), dtype=int32, numpy=
array([[9, 9, 9],
       [9, 9, 9]], dtype=int32)>
```

`tf.fill` (https://www.tensorflow.org/api_docs/python/tf/fill) evaluates at graph runtime and supports dynamic shapes based on other runtime `tf.Tensors`, unlike `tf.constant(value, shape=dims)` (https://www.tensorflow.org/api_docs/python/tf/constant), which embeds the value as a `Const` node.

Args	
`dims`	A 1-D sequence of non-negative numbers. Represents the shape of the output `tf.Tensor` (https://www.tensorflow.org/api_docs/python/tf/Tensor). Entries should be of type: `int32`, `int64`.
`value`	A value to fill the returned `tf.Tensor` (https://www.tensorflow.org/api_docs/python/tf/Tensor).
`name`	Optional string. The name of the output `tf.Tensor` (https://www.tensorflow.org/api_docs/python/tf/Tensor).
`layout`	Optional, `tf.experimental.dtensor.Layout` (https://www.tensorflow.org/api_docs/python/tf/experimental/dtensor/Layout). If provided, the result is a DTensor (https://www.tensorflow.org/guide/dtensor_overview) with the provided layout.

Returns
A `tf.Tensor` (https://www.tensorflow.org/api_docs/python/tf/Tensor) with shape `dims` and the same dtype as `value`.

Raises	
`InvalidArgumentError`	`dims` contains negative entries.
`NotFoundError`	`dims` contains non-integer entries.

numpy compatibility

Similar to `np.full`. In `numpy`, more parameters are supported. Passing a number argument as the shape (`np.full(5, value)`) is valid in `numpy` for specifying a 1-D shaped result, while TensorFlow does not support this syntax.

Last updated 2024-04-26 UTC.